

**MORGANTOWN, WV, USA**WMO: **724176**Lat: **39.643N**Lon: **79.916W**Elev: **378**StdP: **96.87**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **13736****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.1      | -11.3 | -20.0                         | 0.7 | -12.2 | -17.3 | 0.9 | -9.5 | 8.0                   | 7.6  | 7.2 | 5.2  | 2.2                   | 230  | 0.417 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.3                   | 31.8            | 22.2 | 30.6 | 21.7 | 29.4 | 21.1 | 24.0                | 29.2 | 23.3 | 28.0 | 22.6 | 27.1 | 2.7                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.5                            | 18.0 | 26.2 | 21.8 | 17.3 | 25.6 | 21.2 | 16.6 | 25.0 | 73.8          | 29.4 | 70.9 | 28.1 | 68.2 | 26.9 | 27.3              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |  |  |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|--|--|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |  |  |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         |  |  |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |  |  |
| (4) <b>6.8</b>    | <b>5.7</b> | <b>5.3</b> | <b>-17.9</b>               | <b>33.5</b> | <b>3.4</b>         | <b>1.7</b> | <b>-20.3</b>                                       | <b>34.8</b> | <b>-22.3</b> | <b>35.8</b> | <b>-24.2</b> | <b>36.8</b> | <b>-26.7</b> | <b>38.0</b> |  |  |
| (5)               |            |            | <b>-18.3</b>               | <b>25.2</b> | <b>3.3</b>         | <b>0.9</b> | <b>-20.7</b>                                       | <b>25.9</b> | <b>-22.7</b> | <b>26.4</b> | <b>-24.5</b> | <b>26.9</b> | <b>-27.0</b> | <b>27.6</b> |  |  |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.2   | -0.1  | 1.5   | 5.9   | 12.3  | 17.0  | 21.3  | 23.2  | 22.6  | 19.3  | 13.1  | 7.2   | 2.7   |
|                                                                                  | DBStd       | 9.62   | 7.10  | 6.37  | 6.19  | 5.43  | 4.50  | 3.19  | 2.61  | 2.55  | 3.78  | 4.90  | 5.38  | 5.87  |
|                                                                                  | HDD10.0     | 1148   | 324   | 246   | 160   | 37    | 3     | 0     | 0     | 0     | 0     | 25    | 116   | 237   |
|                                                                                  | HDD18.3     | 2744   | 573   | 471   | 388   | 191   | 80    | 11    | 1     | 2     | 34    | 174   | 334   | 486   |
|                                                                                  | CDD10.0     | 1958   | 9     | 8     | 32    | 106   | 221   | 338   | 409   | 392   | 278   | 121   | 33    | 11    |
|                                                                                  | CDD18.3     | 513    | 0     | 0     | 2     | 10    | 40    | 100   | 152   | 135   | 62    | 12    | 1     | 0     |
|                                                                                  | CDH23.3     | 4037   | 0     | 1     | 12    | 106   | 353   | 783   | 1207  | 1015  | 468   | 89    | 4     | 0     |
|                                                                                  | CDH26.7     | 1167   | 0     | 0     | 1     | 18    | 85    | 224   | 392   | 303   | 130   | 13    | 0     | 0     |
| Wind                                                                             | WSAvg       | 2.2    | 2.7   | 2.7   | 2.7   | 2.7   | 2.2   | 2.0   | 1.8   | 1.6   | 1.7   | 2.0   | 2.3   | 2.5   |
| Precipitation                                                                    | PrecAvg     | 1045   | 66    | 61    | 95    | 87    | 103   | 101   | 109   | 99    | 89    | 67    | 83    | 78    |
|                                                                                  | PrecMax     | 1547   | 156   | 134   | 185   | 151   | 227   | 239   | 304   | 247   | 163   | 186   | 276   | 184   |
|                                                                                  | PrecMin     | 750    | 16    | 16    | 26    | 29    | 32    | 22    | 35    | 26    | 20    | 6     | 22    | 16    |
|                                                                                  | PrecStd     | 166    | 33    | 33    | 38    | 31    | 47    | 50    | 55    | 48    | 33    | 37    | 44    | 36    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 19.5  | 20.8  | 25.3  | 28.9  | 30.8  | 32.3  | 33.5  | 32.8  | 32.1  | 28.4  | 23.4  | 19.8  |
|                                                                                  |             | MCWB   | 13.6  | 12.7  | 15.0  | 16.6  | 20.1  | 22.4  | 23.6  | 22.9  | 21.0  | 19.7  | 15.5  | 14.6  |
|                                                                                  | 2%          | DB     | 16.6  | 17.4  | 22.1  | 26.3  | 28.9  | 30.6  | 31.8  | 31.3  | 30.0  | 25.8  | 20.6  | 16.9  |
|                                                                                  |             | MCWB   | 12.7  | 11.0  | 13.3  | 15.4  | 19.2  | 21.7  | 22.7  | 22.4  | 20.6  | 18.1  | 13.9  | 12.5  |
|                                                                                  | 5%          | DB     | 13.7  | 14.3  | 19.2  | 24.0  | 27.3  | 29.2  | 30.3  | 29.7  | 28.0  | 23.6  | 18.3  | 14.5  |
|                                                                                  |             | MCWB   | 10.9  | 9.8   | 11.9  | 14.6  | 18.8  | 21.2  | 22.0  | 21.8  | 20.1  | 16.8  | 12.6  | 10.8  |
|                                                                                  | 10%         | DB     | 10.6  | 11.4  | 16.1  | 21.6  | 25.2  | 27.8  | 29.0  | 28.4  | 26.2  | 21.4  | 16.2  | 12.2  |
|                                                                                  |             | MCWB   | 7.5   | 7.3   | 10.4  | 13.5  | 17.9  | 20.5  | 21.5  | 21.2  | 19.3  | 15.9  | 11.6  | 9.0   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.1  | 14.7  | 16.5  | 18.7  | 22.4  | 24.1  | 25.2  | 25.1  | 23.4  | 21.4  | 17.0  | 15.2  |
|                                                                                  |             | MCDB   | 18.2  | 18.9  | 23.6  | 25.7  | 28.0  | 29.8  | 30.6  | 30.0  | 28.3  | 26.1  | 21.1  | 18.4  |
|                                                                                  | 2%          | WB     | 13.1  | 12.6  | 14.5  | 17.0  | 20.8  | 23.0  | 24.1  | 23.9  | 22.3  | 19.8  | 15.2  | 13.4  |
|                                                                                  |             | MCDB   | 15.5  | 15.8  | 19.4  | 23.0  | 26.5  | 28.0  | 29.6  | 28.6  | 26.9  | 24.0  | 19.3  | 16.1  |
|                                                                                  | 5%          | WB     | 11.1  | 10.2  | 13.1  | 15.9  | 19.9  | 22.2  | 23.3  | 23.1  | 21.3  | 18.3  | 13.6  | 11.5  |
|                                                                                  |             | MCDB   | 13.5  | 13.8  | 17.9  | 21.4  | 25.1  | 27.3  | 28.2  | 27.4  | 25.5  | 21.9  | 17.0  | 13.8  |
|                                                                                  | 10%         | WB     | 7.7   | 7.7   | 11.1  | 14.8  | 18.8  | 21.5  | 22.6  | 22.4  | 20.5  | 16.6  | 12.0  | 9.0   |
|                                                                                  |             | MCDB   | 10.1  | 11.1  | 15.6  | 20.1  | 23.6  | 26.3  | 27.0  | 26.5  | 24.4  | 20.6  | 15.4  | 11.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.3   | 9.1   | 10.3  | 11.9  | 11.2  | 10.6  | 10.3  | 10.4  | 10.8  | 10.8  | 9.4   | 8.0   |
|                                                                                  |             | MCDBR  | 10.7  | 13.1  | 14.4  | 14.7  | 13.7  | 12.2  | 11.8  | 11.9  | 13.1  | 13.5  | 12.8  | 11.6  |
|                                                                                  |             | MCWBR  | 8.1   | 9.3   | 8.3   | 7.2   | 6.3   | 5.1   | 4.6   | 4.7   | 5.4   | 6.9   | 8.1   | 8.5   |
|                                                                                  | 5% WB       | MCDBR  | 10.2  | 12.1  | 12.1  | 12.0  | 11.3  | 10.6  | 10.0  | 9.8   | 10.3  | 10.8  | 10.6  | 10.5  |
|                                                                                  |             | MCWBR  | 8.2   | 9.8   | 8.2   | 6.8   | 5.6   | 4.9   | 4.4   | 4.5   | 4.9   | 6.1   | 7.8   | 8.7   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.301 | 0.322 | 0.348 | 0.386 | 0.434 | 0.467 | 0.488 | 0.498 | 0.411 | 0.365 | 0.324 | 0.301 |
|                                                                                  | taud        |        | 2.459 | 2.384 | 2.388 | 2.314 | 2.187 | 2.127 | 2.087 | 2.031 | 2.267 | 2.398 | 2.488 | 2.496 |
|                                                                                  | Ebn at Noon |        | 871   | 897   | 907   | 889   | 849   | 818   | 796   | 775   | 828   | 835   | 836   | 843   |
|                                                                                  | Edh at Noon |        | 79    | 98    | 109   | 125   | 145   | 154   | 160   | 164   | 122   | 96    | 77    | 71    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1.60  | 2.42  | 3.45  | 4.60  | 5.25  | 5.78  | 5.65  | 5.18  | 4.29  | 2.98  | 2.04  | 1.40  |
|                                                                                  | RadStd      |        | 0.18  | 0.38  | 0.29  | 0.41  | 0.45  | 0.44  | 0.34  | 0.30  | 0.48  | 0.35  | 0.25  | 0.15  |